

$$= 4 (I_0)^2 R = 4 P.$$

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D

To find the p.d. V_p in the primary coil,

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$V_p = \frac{N_p}{N_s} \times V_s = \frac{200}{1000} \times 80V = 16V$$

Power dissipated on primary coil = power dissipated on secondary coil

$$I_p V_p = \frac{V_s^2}{R}$$

$$I_p (16V) = \frac{(80V)^2}{100\Omega}$$

$$I_p = 4.0A$$

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B

de Broglie's equation is